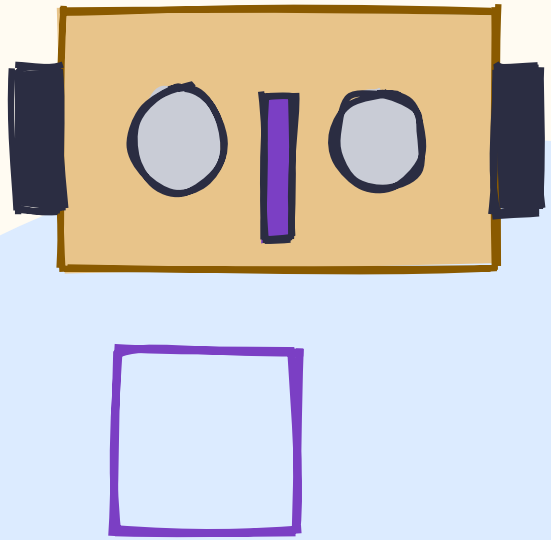
Young Innovators · Class 4

Stepper Motor Drawing Plotter

Build a robot that draws exactly what your code says

Stepper motors | Calibration | Functions



Did you know?
Your school printer, 3D printers and CNC cutting machines all move with stepper motors.

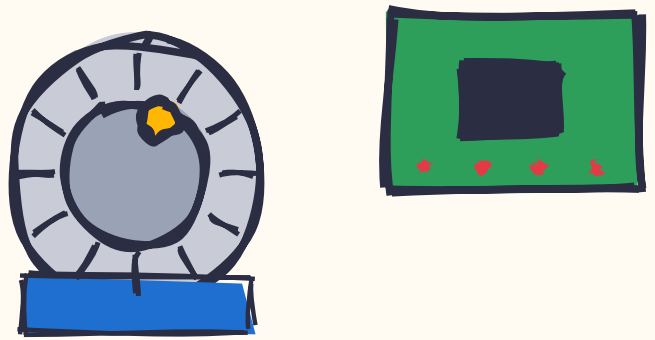
LEARN

Let's think...

How does a printer know where to put the ink?

Exactly 0.1 mm

A printer head stops in exactly the right place, thousands of times a page. A normal DC motor just spins until you switch it off. So how does a machine move an **exact** distance, with no camera checking?



28BYJ-48

Very fast eyes?

A timer?

Counting steps!

Today: you build a two-wheeled robot that **draws shapes** with a pen, using two stepper motors.

LEARN

Learning objectives

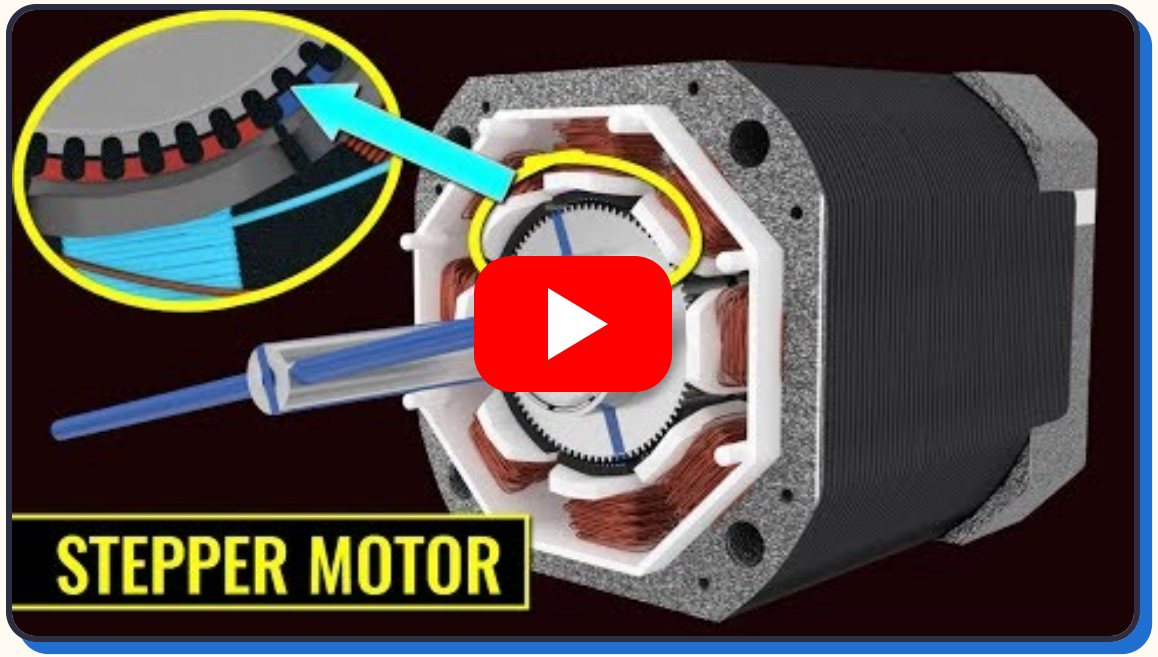
By the end of this class I can...

1	**Steps:** explain how a **stepper motor** turns in tiny, exact steps	**Proof:** my motor turns exactly one full turn (2048 steps) and stops
2	**Calibrate:** work out **steps per mm** and **steps per degree** for my robot	**Proof:** my robot drives 100 mm within 5 mm
3	**Draw with code:** write **functions** that draw shapes from straight lines and turns	**Proof:** my robot draws a closed square
4	**Improve:** measure my errors and **fix** them by changing one number	**Proof:** my second square closes better than my first

LEARN

Objective 1 · Steps · video

Video time!



How does a Stepper Motor work?

Sabin Civil Engineering · about 6 minutes · youtu.be/eyqwLiowZiU

Watch for...

1. Why does the rotor **stop** exactly where the coils pull it?
2. What is the difference between **full step** and **half step**?
3. Name **two machines** in the video that use steppers.

Scan to watch

After the video: Turn your pencil in tiny clicks, like a clock's second hand. Count: how many clicks for one full turn?

Objective 1 · Steps

A stepper moves one click at a time

The idea

Inside are **4 coils**. Switch them on **one after another** and the magnet rotor jumps forward one small step each time.

The 28BYJ-48 has a gearbox inside, so it needs **2048 steps** for one full turn. One step is only about **0.18°**!

Open loop
No sensor is needed. The Arduino just counts the steps it sent. Engineers call this OPEN-LOOP control.

Example: a clock's second hand



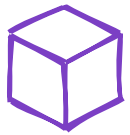
60 ticks

= 1 turn

A second hand does not glide: it **ticks**. 60 ticks = one full turn. If you count the ticks, you know exactly where it is, without looking. A stepper motor is the same, just with 2048 ticks.

Real world

Steppers make machines precise



3D printers
Move the nozzle in X, Y and Z, step by step



Printers & scanners
Slide the print head across the paper



CNC machines
Cut wood and metal parts to 0.01 mm



Camera lenses
Focus by moving glass a tiny exact amount

? If a stepper is told to move but something blocks it, will the Arduino know? Why could that be a problem?

Missed steps
If the motor is blocked, it skips steps and the machine gets lost. Good machines avoid overloading it.

LEARN

Objective 2 · Calibrate

Calibrate: turn steps into millimetres

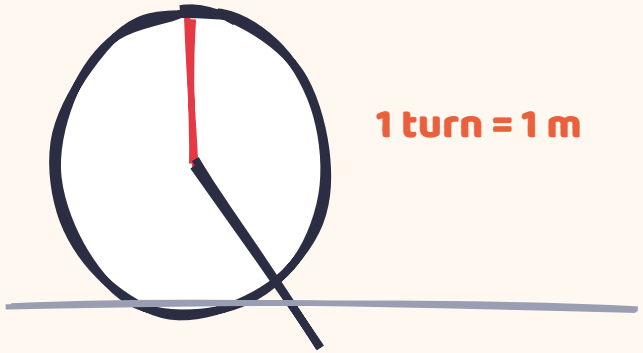
The idea

One wheel turn moves the robot one **circumference**: **π × diameter**.
Wheel 60 mm → 3.14 × 60 = **188 mm** per 2048 steps.
steps per mm = 2048 ÷ 188 ≈ 10.9

For turning on the spot, each wheel travels part of a circle whose diameter is the **wheelbase** (distance between wheels).

Measure!
Measure YOUR wheel. A 2 mm difference in diameter makes a 100 mm line about 3 mm wrong.

Example: a trundle wheel



A trundle wheel in PE clicks once every metre. Count the clicks and you know the distance, because the wheel's circumference is exactly 1 m. Our robot counts steps instead of clicks.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human plotter
2. Our plotter kit (per group)
3. Wire two steppers and a servo
4. One exact turn
5. Functions that draw
6. Build the robot body
7. Test, measure, fix

MAKE

Class game

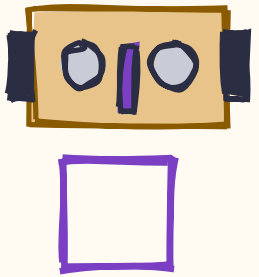
Human plotter

How to play

1. One student is the **robot**, holding a marker over a big sheet of paper on the floor.
2. They may only obey: **FORWARD n** (n heel-to-toe steps), **LEFT 90**, **RIGHT 90**, **PEN UP**, **PEN DOWN**.
3. Your team writes a program to draw a **house** (square + triangle roof).
4. Swap programs with another team and run theirs.

Then discuss

Did the house close neatly? Whose 'steps' were longer? What would you need to measure to make it exact?



Big idea

Machines need commands they can count, plus a way to turn those counts into real distances.

Build it

Our plotter kit (per group)



Arduino Uno
Counts and sends the steps



2 × 28BYJ-48 + ULN2003
Left and right wheels



Servo SG90 + marker
Lifts the pen up and down



4×AA pack
Powers the motors



Cardboard + tape
Chassis and wheels



Laptop
Arduino IDE

Wheels: two bottle caps or cardboard discs about 60 mm across, taped firmly to the motor shafts. A paper clip or bead at the front as a skid.

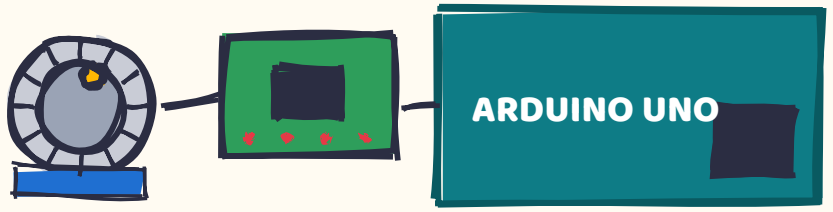
Power
Steppers draw too much current for USB alone. Power the ULN2003 boards from the battery pack and join all the GNDs.

MAKE

Objective 1 · Steps · Build step 1 of 3

Wire two steppers and a servo

Part	Connect to
Left ULN2003 IN1, IN2, IN3, IN4	pins 2, 3, 4, 5
Right ULN2003 IN1, IN2, IN3, IN4	pins 6, 7, 8, 9
Both ULN2003 + and –	battery + and –
Servo signal (orange)	pin 10
Servo red / brown	5V / GND
Battery –	Arduino GND (common ground!)



Common ground

If the battery – is not joined to Arduino GND, the motors just buzz and shake.

MAKE

Objective 1 · Steps · test it

One exact turn

```
#include <Stepper.h>
const int SPR = 2048;              // steps per revolution
// order IN1, IN3, IN2, IN4 for this motor
Stepper left(SPR, 2, 4, 3, 5);

void setup() {
  left.setSpeed(12);               // rpm (max about 15)
}

void loop() {
  left.step(SPR);                  // one full turn forward
  delay(1000);
  left.step(-SPR / 4);             // a quarter turn back
  delay(1000);
}
```

Mark it

Put a tape flag on the shaft. Does it come back to the **same place** after one turn?

Pin order!

Using 2, 3, 4, 5 in that order makes the motor just vibrate. The library needs **IN1, IN3, IN2, IN4**.

Try

Change 12 to 20 rpm. What happens? (It may stall and skip.)

MAKE

Objective 3 · Draw with code · code it

Functions that draw

```
#include <Stepper.h>
#include <Servo.h>
Stepper L(2048, 2, 4, 3, 5), R(2048, 6, 8, 7, 9);
Servo pen;
float STEPS_PER_MM  = 10.9;   // calibrate!
float STEPS_PER_DEG = 9.5;    // calibrate!

void drive(float mm) {
  long n = mm * STEPS_PER_MM;
  for (long i = 0; i < n; i++) { L.step(1); R.step(-1); }
}
void turn(float deg) {            // turn on the spot
  long n = deg * STEPS_PER_DEG;
  for (long i = 0; i < n; i++) { L.step(1); R.step(1); }
}
void penDown() { pen.write(40); delay(300); }
void penUp()   { pen.write(90); delay(300); }

void setup() {
  L.setSpeed(12); R.setSpeed(12); pen.attach(10);
  penDown();
  for (int s = 0; s < 4; s++) { drive(80); turn(90); }
  penUp();
}
void loop() {}
```

Why R.step(-1)?

The right motor faces the other way. To roll forward together it must spin the **opposite** direction.

Functions

drive() and **turn()** hide the maths. Now drawing a square is just 4 × (drive, turn).

Pen

Adjust 40 and 90 so the marker just touches the paper when down.

MAKE

Objective 3 · Build step 2 of 3

Build the robot body

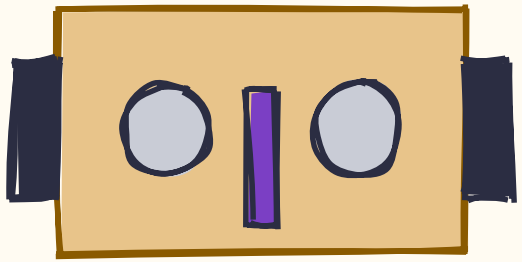
a Tape the two steppers to the **underside** of a cardboard base, shafts pointing outwards.

b Push the wheels on the shafts and wrap the edges with tape or a rubber band for grip.

c Cut a hole in the middle, **between the wheels**, for the marker.

d Tape the servo next to the hole so its arm can lift the marker.

e Add a smooth skid at the front so the robot does not tip.



Check: The robot balances on 2 wheels + skid, and the pen sits exactly between the wheels.

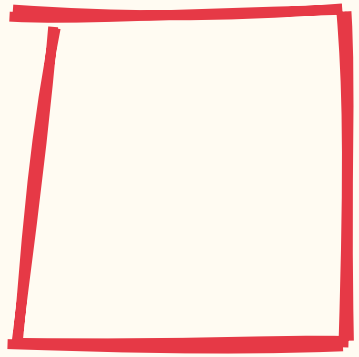
Pen not between the wheels? Every turn will draw a curve instead of a sharp corner.

MAKE

Objective 4 · Improve · Build step 3 of 3

Test, measure, fix

Command	Should be	We measured	New value
drive(100)	100 mm		
turn(360)	360°		
Square 80 mm	closes		
After fixing	closes		



gap = error

Fix with a ratio

Measured 94 mm instead of 100? New STEPS_PER_MM = 10.9 × 100 ÷ 94 = **11.6**. Same idea for STEPS_PER_DEG.

One change at a time

Fix distance first, then turning. Change both together and you won't know which helped.

Test like an engineer

Troubleshooting

Problem	Likely cause	Fix
Motor only buzzes	wrong pin order or weak battery	use IN1, IN3, IN2, IN4
Robot drives in a circle	one motor reversed	flip the sign of R.step
Corners are rounded	pen not between wheels	move the pen hole
Lines are dotted	pen too high	lower the down angle
Square does not close	calibration	adjust the two numbers

Debug checklist

Wheels slipping? Draw on paper, not a shiny desk, and add tape tyres.

Slow down to 8 rpm while testing. Slower steppers are stronger and skip fewer steps.

MAKE

Level up!

Plotter challenges

Choose your challenge



Star: draw a 5-pointed star. Hint: turn 144° each time.



Polygon function: polygon(sides, length) draws any regular shape.



Your initials: plan the letters on squared paper, then code them.



Spiral: drive a little further after each turn.

Think like an engineer

A circle is a polygon with many sides.
How many sides before it looks round?

Boss level

Write circle(radius) using drive() and turn() only.

Objective check

Can I really do it? Prove it.

01 Why does a stepper not need a sensor to know where it is?

02 Show your calculation for steps per mm. What wheel diameter did you measure?

03 Show your square. Point to drive() and turn() in your code.

04 Which number did you change to make the square close, and by how much?

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

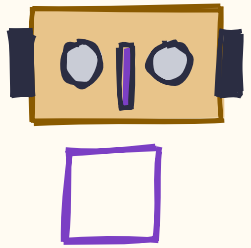
Teacher: Gallery walk: lay all the squares side by side. Whose closes best? Ask them what they changed.

Show what you know

What did we learn?

- [x] Steppers move in exact, countable steps (2048 per turn here)
- [x] Distance = steps ÷ steps per mm (from π × diameter)
- [x] Functions like drive() and turn() make drawing simple
- [x] Measure the error, fix one number, test again
- [x] Printers, 3D printers and CNC machines work the same way

At home
Find 3 things at home that move in exact steps or clicks.



Exit ticket: the robot drew a 90 mm line instead of 100 mm. What do you change, and to what?

Next class: Obstacle-Avoiding Robot with Scanning Sonar!